

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Understanding design criteria with a more logical and creative approach, so when reviewing AI generated code, an engineer can see if it actually solves a problem for example there may still be security issues with such an AI generated application or redundancies or code inefficiency.
2. Security details, backend, how the system actually works
3. An engineer would still need to know how to code in order to review whether or not an AI is hallucinating code or variables or even functions.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	A calculator is usually a basic starter application many students starting out in software make.
B	Yes	Yes	Affects a lot of stakeholders, is wider scaled in terms that if it is so huge, it must be secure to meet needs like security, proper system functions, longevity, etc.
C	Yes	No	Programming because

			although it may have generated an appointment setting application, but it cannot be full proof without proper backend and frontend design. Program alone does not mean a program is secure.
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Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Admin staff	Easy navigational system, streamlining schedule and booking management as well as patient details
IT support staff	Maintaining systems
Practitioners	Access to patient info, patient type, appointment type by need
Patients	Easy booking, retention reminders, accurate personal details (can change when needed)

Task 2 - Identify current problems

1. Spreadsheets can be hard to maintain without correct assigned roles or with extensive data that is not easily manageable
2. Manual scheduling
3. No central portal
4. Paper records are or can be unreliable and harder to maintain.

Task 3 - Ask client questions

1. Are there any current clashes with booking times like duplicate patient records or appointments?
2. Are appointments booked by admin staff over the phone with patients or its flexible?
3. What are your current security flaws/concerns
4. Can you be more specific on the design criteria?
5. Any other features you expect in the final or during the development of the system?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes	Yes	Accept
Facial recognition login	No – patient privacy compliance	No	Reject
AI diagnosis recommendations	No – can be false and inaccurate	No	Reject
Patient search	Yes – helps in querying patients for records and reference	Yes (assumed)	Accept
Online payment	Not mentioned, but useful	Yes	Accept
Practitioner schedule view	Yes	Yes	Accept
Insurance processing	No, but supports patients	Maybe	Consider, consult client
Treatment-plan generation	No – only by practitioner, ai is excluded	No	Reject

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

- Validating code and ensuring security checks in compliance with industry standards is a must to ensure all stakeholders are confident and the system is compliant.